AI Clusters

From Training to Tokens: Revolutionizing AI Inference with In-Network Compute

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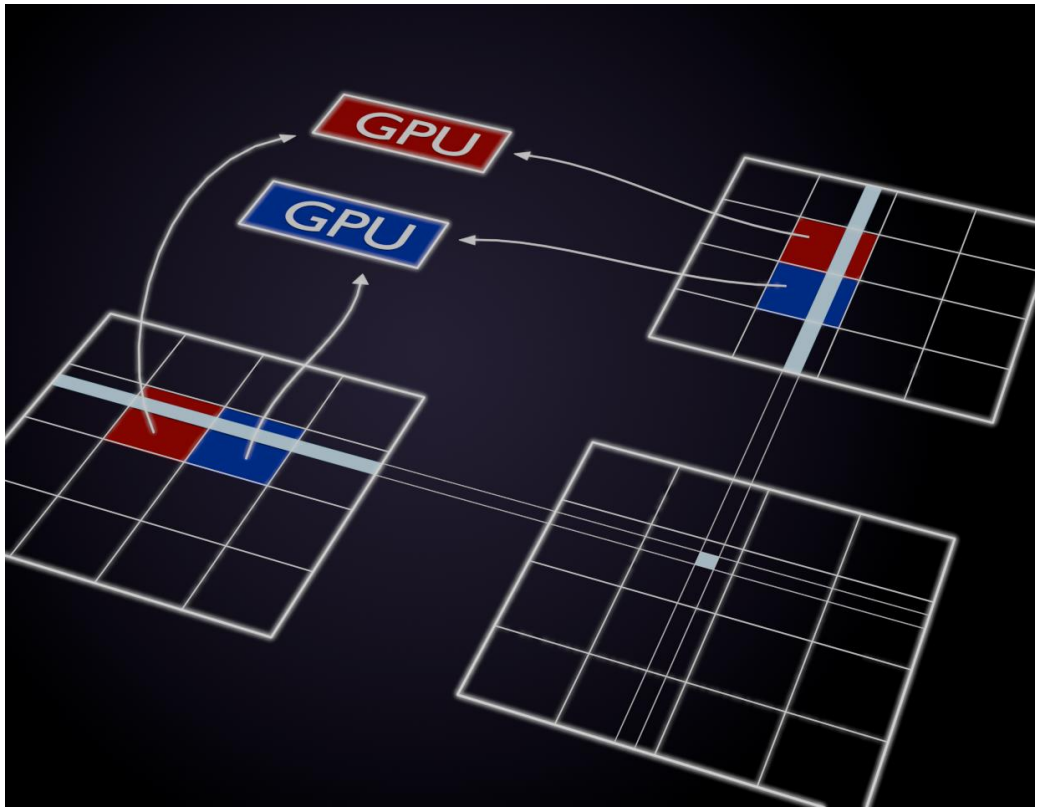
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Outline

1. Patterns in AI traffic: Scale-up, Inference, Mixture-of-Experts (MoE)
2. Opportunities: In-Network Compute (INC) to alleviate AI traffic pain points
3. INC in PCIe®
4. INC everywhere else
5. Call to action: Enhance your collectives libraries

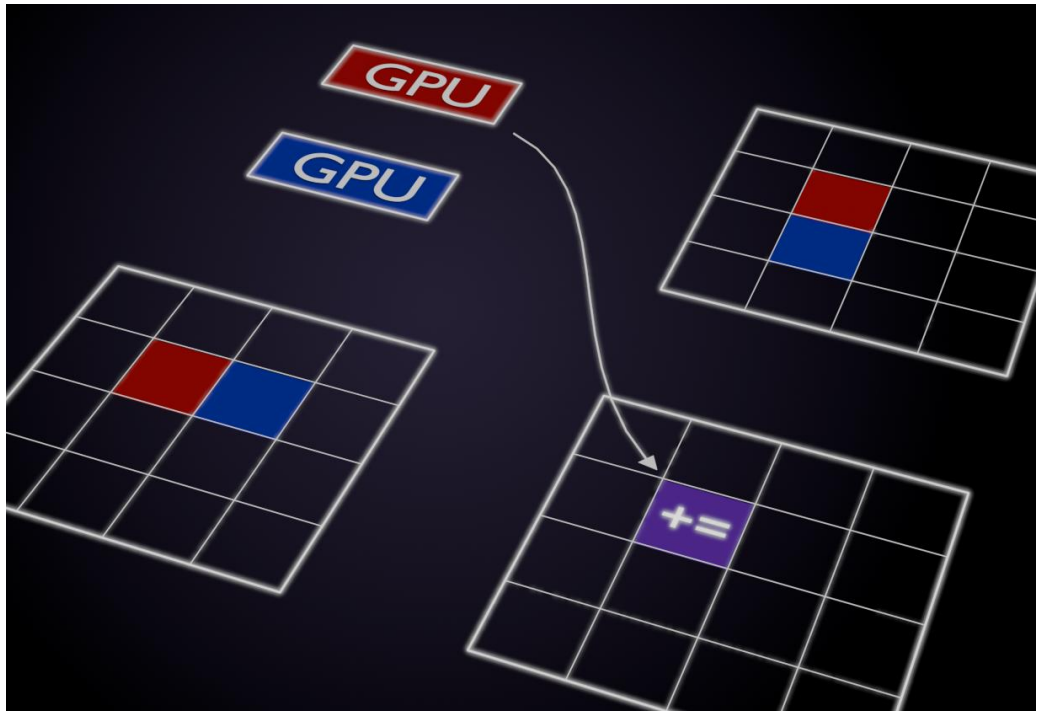
AI Traffic Patterns and Scale-up Network

- AI poses unique challenges to the fabric:
  - Multiple accelerators need to participate in **collective** computation
- Critical collective:
  - Distributed GEMM (**frequent**; bandwidth-hungry; latency-sensitive)
- Critical fabric design requirement:
  - Provide a **bandwidth-efficient** solution with low total **latency**

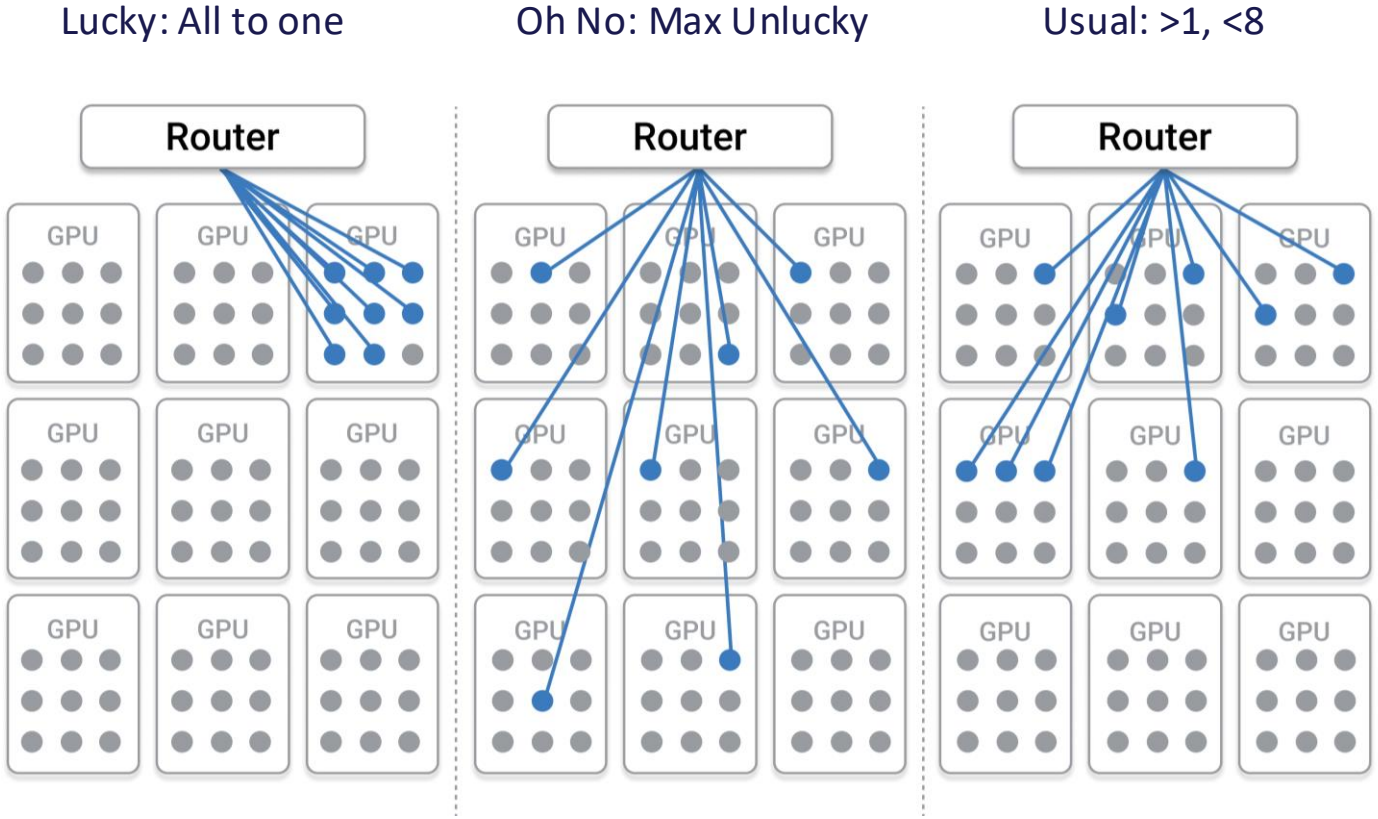


AI Traffic Patterns and Scale-up Network

- Distributed GEMM decomposes into **collective operations**:
  - All Reduce (AR)
  - Reduce Scatter (RS)
  - All Gather (AG)
- Same collectives are useful for other parts of the model data flow such as MoE expert recruitment



AI Traffic Patterns and Scale-up Network



- MoE brings a new motivation for dynamic multicast grouping
- Per-token decision during expert routing
- Worthwhile to augment the performance

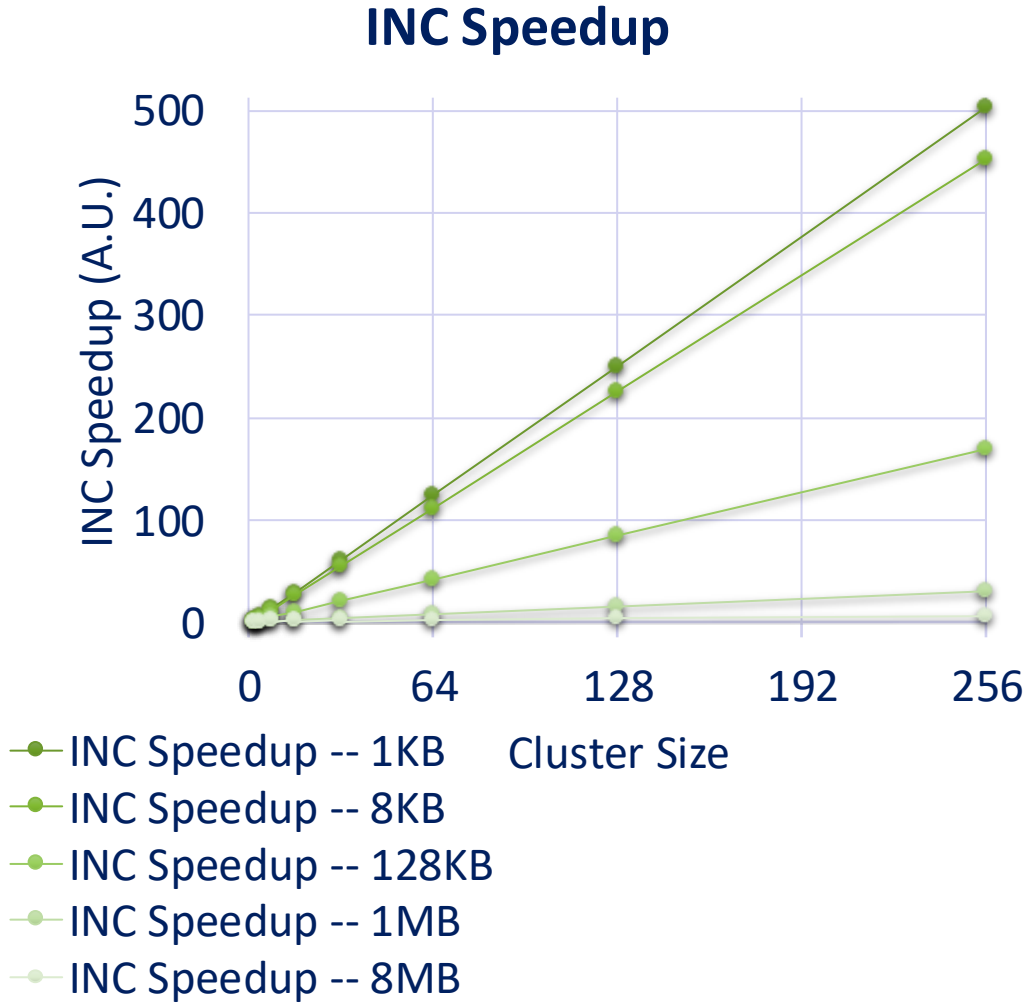
AI Traffic Patterns and Scale-up Network

Key takeaways from this section:

- Two inference requirements for the fabric
  - Distributed GEMM acceleration
  - Dynamic communication patterns (MoE)
- Criticality of latency
- Need for topology-aware EP placement for operands participating in collectives (prefer minimum hops)

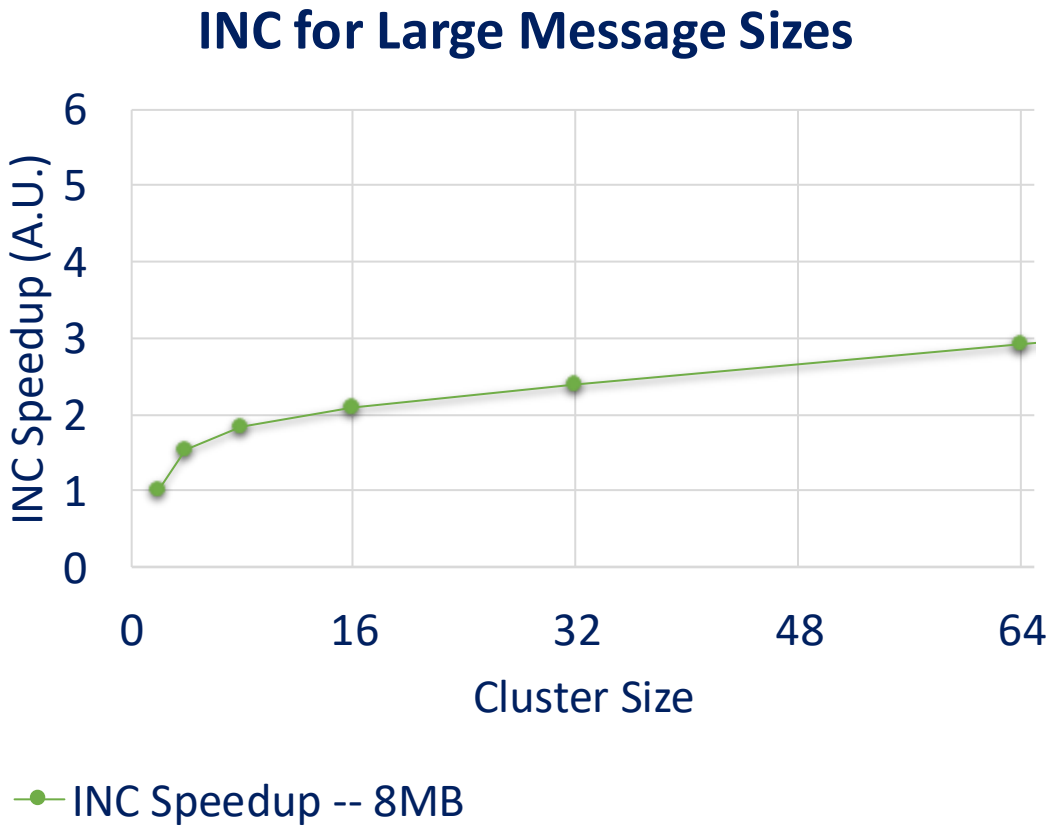
Inference is the Main Winner

- INC speed up of **small message sizes** linearly grow with the size of the cluster.
- Inference message sizes in **hundreds of KB** show large speedup potentials.



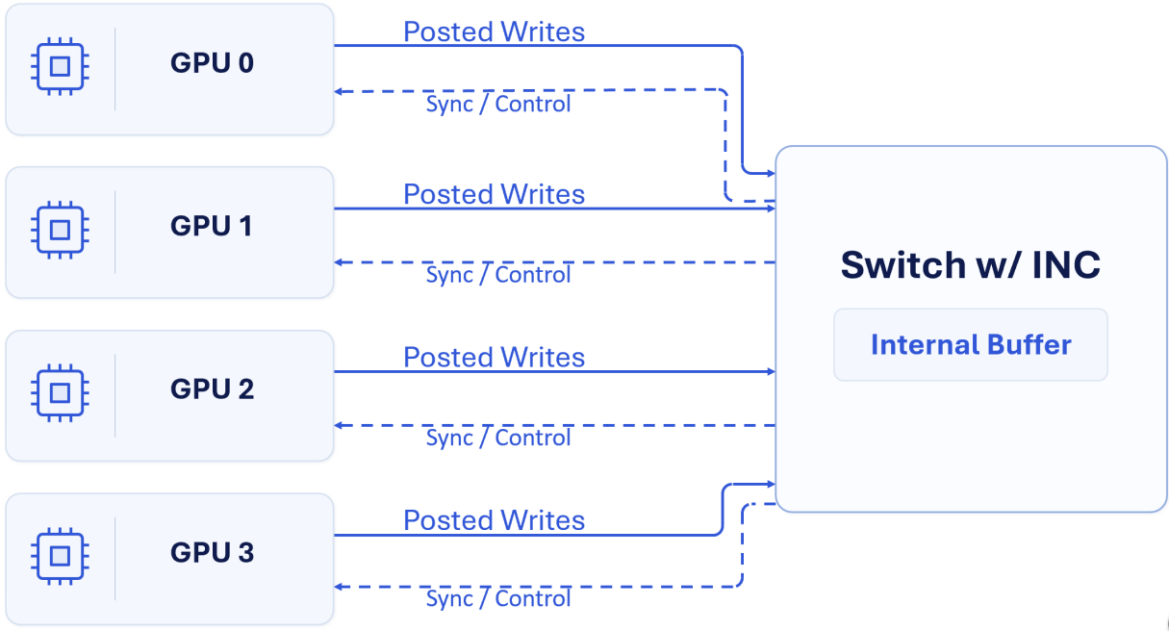
Training Benefits Too!

- Training message sizes in **multiples of MB** show non-negligible speedup potentials, as well.



How to Make INC Work

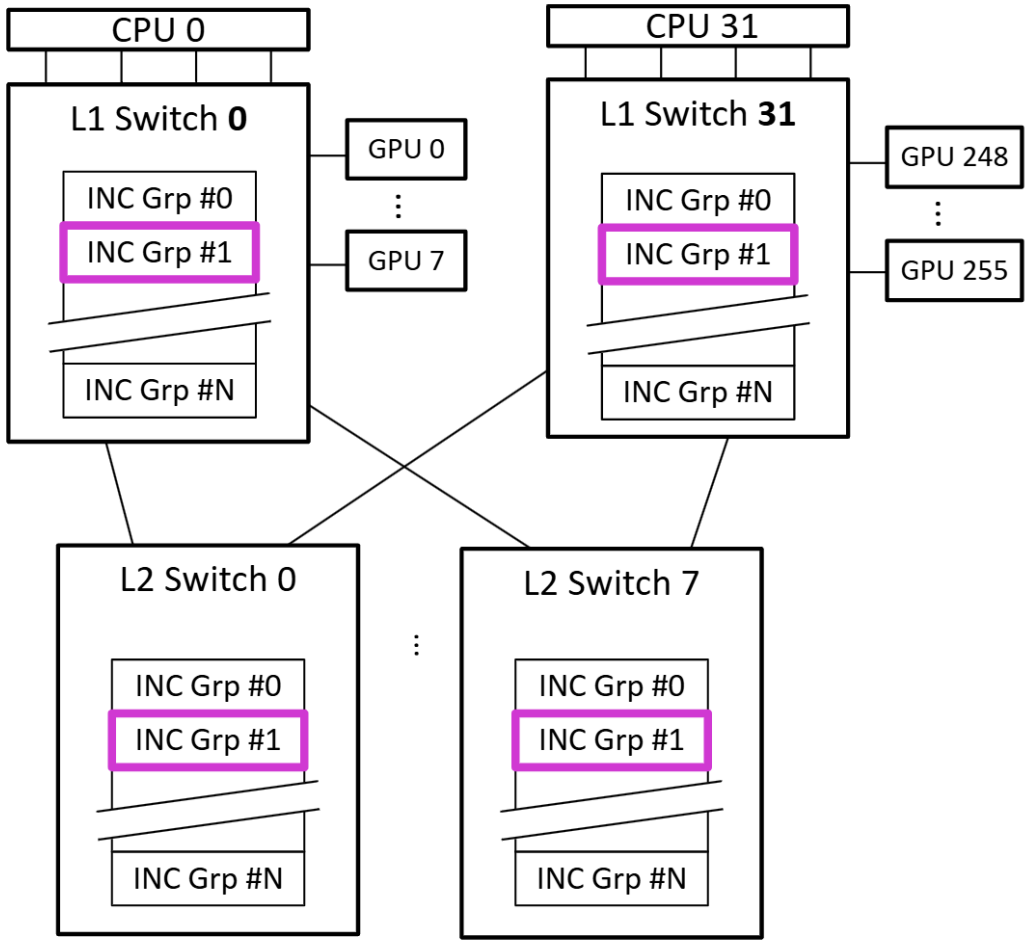
- Enough space in the switch is required, while the partial result in the switch is waiting for the copies from all the GPUs.
- Latency and Synchronization are the keys!
  - Pull model: Switch generates read request to GPUs
  - Push Model: GPUs generate posted writes to the switch.



INC in PCIe Fabric

- Dynamic multicast grouping: lightweight config concept
- Diagram: Example AG in 256-node PCIe Fabric Routing diagram with MC

Ample buffer size and context count are critical HW features, ready to take advantage of to optimize AllGather, ReduceScatter and AllReduce, management?



INC Everywhere Else



- INC offers a linear speedup for inference collectives as the cluster grows.
- Today INC is available in almost all of the scale-up protocols:
  - NVLink
  - Ethernet/UEC
  - PCIe
    - Light-weight configuration
    - Flexible addressing
    - Efficient synchronization

Call to Action

- Making INC portable and open will be a community effort:
  - We've offered a solution for the synchronization path and a lightweight configuration method – send us your feedback
  - Prepare library hooks and integrations to enable INC operation in the fabric
  - Consider push and pull model both – focus on total operation **latency**
  - Provide topology-aware data placement to take advantage of **fewest-hop** collectives
- Meet with Astera Labs experts: Booth B34

Thank You!

